AN

INTRODUCTION

TO

CALCULUS

BY RENIS,

AUTHOR OF DATA STRUCTURES AND ALGORITHMS.

This book is the same as the last seria is juast a way of taking notes and to learn calculus

Published by:

Just me taking notes with LaTeX

2026.

CONTENT

Chapter 0 Highlights of Calculus

0.1 Distance and Speed // Height and Slope 1

0.2 The Changing Slope of $y = x^2$ and $y = x^n$ 9

0.3 The Exponential $y = e^x$ 15

0.4 Video Summaries and Practice Problems 23

0.5 Graphs and Graphing Calculators 45

Chapter 1 Introduction to Calculus

1.1 Velocity and Distance 51

1.2 Calculus Without Limits 59

1.3 The Velocity at an Instant 67

1.4 Circular Motion 73

1.5 A Review of Trigonometry 80

1.6 A Thousand Points of Light 85

Chapter 2 Derivatives

2.1 The Derivative of a Function 87

2.2 Powers and Polynomials 94

2.3 The Slope and the Tangent Line 102

2.4 The Derivative of the Sine and Cosine 109

2.5 The Product and Quotient and Power Rules 116

2.6 Limits 123

2.7 Continuous Functions 131

Chapter 3 Applications of the Derivative

3.1 Linear Approximation 138
3.2 Maximum and Minimum Problems 143
3.3 Second Derivatives: Bending and Acceleration 153
3.4 Graphs 160
3.5 Parabolas, Ellipses, and Hyperbolas 170
3.6 Iterations $x_{n+1} = F(x_n)$ 179
3.7 Newton's Method (and Chaos) 187
3.8 The Mean Value Theorem and l'Hôpital's Rule 197

Chapter 4 Derivatives by the Chain Rule

4.1 The Chain Rule 204
4.2 Implicit Differentiation and Related Rates 211
4.3 Inverse Functions and Their Derivatives 216
4.4 Inverses of Trigonometric Functions 223

Chapter 5 Integrals

5.1 The Idea of the Integral 229
5.2 Antiderivatives 234
5.3 Summation versus Integration 240
5.4 Indefinite Integrals and Substitutions 249
5.5 The Definite Integral 254
5.6 Properties of the Integral and Average Value 260
5.7 The Fundamental Theorem and Its Applications 267
5.8 Numerical Integration 275

Chapter 6 Exponentials and Logarithms

6.1 An Overview 284
6.2 The Exponential e^x 292
6.3 Growth and Decay in Science and Economics 299
6.4 Logarithms 310
6.5 Separable Equations Including the Logistic Equation 317
6.6 Powers Instead of Exponentials 326
6.7 Hyperbolic Functions 336

Chapter 7 Techniques of Integration

7.1 Integration by Parts 342
7.2 Trigonometric Integrals 348
7.3 Trigonometric Substitutions 355
7.4 Partial Fractions 362
7.5 Improper Integrals 367

Chapter 8 Applications of the Integral

8.1 Areas and Volumes by Slices 373
8.2 Length of a Plane Curve 383
8.3 Area of a Surface of Revolution 388
8.4 Probability and Calculus 391
8.5 Masses and Moments 399
8.6 Force, Work, and Energy 406

Chapter 9 Polar Coordinates and Complex Numbers

9.1 Polar Coordinates 412
9.2 Polar Equations and Graphs 416
9.3 Slope, Length, and Area for Polar Curves 421
9.4 Complex Numbers 425

Chapter 10 Infinite Series

10.1 The Geometric Series 433
10.2 Convergence Tests: Positive Series 440
10.3 Convergence Tests: All Series 448
10.4 The Taylor Series for e^x, $\sin x$, and $\cos x$ 452
10.5 Power Series 458

Chapter 11 Vectors and Matrices

11.1 Vectors and Dot Products 466
11.2 Planes and Projections 476
11.3 Cross Products and Determinants 486
11.4 Matrices and Linear Equations 496
11.5 Linear Algebra 507

Chapter 12 Motion Along a Curve

12.1 The Position Vector 517
12.2 Plane Motion: Projectiles and Cycloids 525
12.3 Curvature and Normal Vector 531
12.4 Polar Coordinates and Planetary Motion 537

Chapter 13 Partial Derivatives

13.1 Surfaces and Level Curves 545
13.2 Partial Derivatives 549
13.3 Tangent Planes and Linear Approximations 554
13.4 Directional Derivatives and Gradients 565
13.5 The Chain Rule 574
13.6 Maxima, Minima, and Saddle Points 582
13.7 Constraints and Lagrange Multipliers 592

Chapter 14 Multiple Integrals

14.1 Double Integrals 599
14.2 Change to Better Coordinates 607
14.3 Triple Integrals 616
14.4 Cylindrical and Spherical Coordinates 622

Chapter 15 Vector Calculus

15.1 Vector Fields 631
15.2 Line Integrals 637
15.3 Green's Theorem 646
15.4 Surface Integrals 657
15.5 The Divergence Theorem 667
15.6 Stokes' Theorem and the Curl of F 674

Chapter 16 Mathematics after Calculus

PREFACE

These are just my personal notes; I would like to share my learning journey with you. This is not a formal project, but rather a way for me to master LaTeX and appreciate the beauty of classical typesetting—it takes me back to the era when mathematicians worked solely with pen and paper. For this course, I am using the Calculus book by Gilbert Strang. My goal is not to rewrite Strang's work, but to deepen my understanding by solving every exercise. More importantly, I intend to use this as a roadmap connecting two worlds: the world of software (where I created `avg-dev` *to master low-level programming in C, C++, Rust, Data Structures, Algorithms, and Operating Systems) and the world of Mathematics. I aim to work consistently and realistically, taking the time to truly grasp the concepts—even if it means reading the book twice. Whether this project evolves into* `avg-math` *or* `avg-ai`*, my ultimate goal is to bridge software engineering with Artificial Intelligence, converging at the intersection of hardware (GPU/CUDA). This course will cover Calculus, Algebra, and the mathematical algorithms behind LLMs, building the strong foundation I need to become a capable researcher in the future.*